

	CSCI 556, Fall 2026 Introduction to Cryptography	
	Course Info	Course outline

Course Info:

Course descriptions and objectives	This course features a rigorous introduction to modern Cryptography -- a field that conducts mathematical & algorithmic studies of concepts, methods, and tools for protecting information in computer and communication systems. The course will focus on: • Connection between machine learning and cryptography • Information-theoretic and computational views of security, privacy, and knowledge • Fundamental cryptographic primitives (public-key encryption, digital signatures, pseudo-random number generation, and key agreement, etc) • Basic protocols to guarantee confidentiality and authenticity of data and computation (secret sharing, homomorphic encryption, interactive and zero- knowledge proofs, and multi-party secure computation, digital money, etc) • Computational complexity requirements in cryptography and practical implementation of cryptographic algorithms/protocols We will also cover the relevant number theory, complexity theory, and learning theory. Students in the class are expected to have a reasonable degree of mathematical sophistication, and to be familiar with the basic notions of algorithms and data structures, discrete mathematics, and probability. Prerequisites: CSCI 270 or permission of the instructor. If you have doubts about meeting these prerequisites, please contact the instructor.
Instructor	Professor Shang-Hua Teng Office hours: Monday 12:15(noon)- 1:45 GCS SB9 (B2) or by appointment; Email: shanghua@usc.edu
TA	Sampad Mohanty Email: sbmohant@usc.edu
Textbook	Introduction to Modern Cryptography -- Second Edition Jonathan Katz and Yehuda Lindell, Chapman & Hall/CRC Cryptography and Network Security Series The class will also cover additional material drawn from research papers as well as other books in Theoretical Computer Science. Particularly: A Graduate Course in Applied Cryptography by Dan Boneh and Victor Shoup
Lectures	10:00-11:50am Monday and Wednesday, in room VPD 105
Grades	15%: Class Participation 20%: Quiz 20%: Theoretical Thinking Assignment: Mathematical Formulation (two-page writing and 5-minutes class presentation) 20%: Project: Class Presentation (30 minutes -- team of three) 25%: Project: Term Report (5-page writing, individual)
Integrity	<u>Plagiarism and other anti-intellectual behavior will be dealt with severely. This includes the possibility of failing the course or being expelled from the University.</u>

Course Outline (subject to changes)

% No Class - Labor Day

Lec#	Date	Topic and/or Event	Required Reading	"Fun" Research Reading
1	08/24	Class organization: data, information, and knowledge communication and computation How to model information, knowledge, security, and privacy Perfect Information Security vs Computational Security P vs NP Symmetric ciphers vs Public-Key	Chapters 1-3 Search on the Web or with an AI system. For example: "[google chatgpt] P NP" "[google chatgpt] RSA" "[google chatgpt] Zero Knowledge Proof" "[google chatgpt] MAC" "[google chatgpt] Digital Signature" "[google chatgpt] Homomorphic	Chapter 1.3

		Encryption Indistinguishability and unpredictability Randomization and Interactive Proofs Languages, NLP, ML	Encryption" "[google chatgpt] Differential Privacy" "[google chatgpt] Network Security" "[google chatgpt] Connection between Cryptography and ML"	
2	08/26	Shamir's Secret Sharing Scheme: An Example of Perfect Security Linear Algebra and Number Theory Basic	Handout, Shamir's CACM paper and Chapter 13.3 and Chapter 8	Chapter 1.1
3	08/31	Secret Sharing II: A general Scheme Access Control Monotone Formula Two Basic Primitives general scheme Continue discussion of (perfect) information security	Handout: Benaloh-Leichter, Generalized Secret Sharing and Monotone Functions	Quantum Secret Sharing Efficient Multi-Party Quantum Secret Sharing Schemes
4	09/02	Classic Framework of Secure Communication - - The Setting of Private-Key Encryption Key generation Encryption/Decryption Perfectly-Secret Private-Key Encryption Information and Probability	Chapter 1.2, 1.4, and Chapter 2	
5	09/07	Labor Day (No Class)		
6	09/09	Cryptography vs Machine Learning Cryptanalysis Types of Attacks	Chapter 3	
7	09/14	Computational Approach to Security and Cryptography Complexity Theory Basics Randomness vs Pseudorandomness	Chapter 3.1, 3.2, 3.3	
8	9/16	Modern Framework of Secure Communication Public-key encryption Chosen Plaintext Attacks Number Theory Basics	Chapters, 10.1. 10.2, and Chapter 7, Chapter 8	
9	9/21	RSA Number Theoretic Basics: Chinese Remainder Theorem	Chapter 10.4, Chapter 7, Chapter 11.1, and Chapter 11.5, Chapter 8	
10	9/23	RSA Number Theoretic Basics	Chapter 10.4 and Chapter 8, Chapter 11.5	
11	09/28	Rabin Encryption Scheme Probabilistic Encryption	Chapter 13.5 and Chapter 13.4	
12	09/30	Diffie-Hellman Key Exchange	Chapter 10	
13	10/05	Quiz		
14	10/07	The day before Fall Recess Discrete Logarithms and El Gamal Encryption Scheme	Chapter 11.4.1	
15	10/12	Pseudorandom Generation: Blum-Micali Construction Number Theory Basics	Chapters 7.4-7.8 and Chapter 8	Chapter 5
16	10/14	Pseudorandom number generation		
17	10/19	Indistinguishability and unpredictability Blum-Blum-Shub Pseudorandom generator	Chapters 7.4-7.8 and Chapter 8	Chapter 5
18	10/21	One Way and trapdoor functions, hardcore bits	Chapter 7.1-7.3	
19	10/26	Interactive and Zero Knowledge Proofs	Hangouts	Hangouts
20	10/28	Zero Knowledge Proofs	Hangouts	
21	11/02	Applications of Zero Knowledge Proofs: Multiparty Computation	Hangouts	
22	11/04	Applications of Zero Knowledge Proofs: Multiparty Computation		
23	11/09	Presentation:		
24	11/11	No Class - Veterans Day		
25	11/16	Presentation:		
26	11/18	Presentation:		
27	11/23	the lecture before thanksgiving break Presentation		

28	11/25	Thanksgiving (No Class)		
29	11/30	Presentation		
30	12/02	Presentation		